

Memory-Efficient Dataflow Inference for Deep CNNs on FPGA

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Abstract—Custom dataflow Convolutional Neural Network (CNN) inference accelerators on FPGA are tailored to a specific CNN topology and store parameters in On-Chip Memory (OCM), resulting in high energy efficiency and low inference latency. However, in these accelerators the shapes of parameter memories are dictated by throughput constraints and do not map well to the underlying OCM, which becomes an implementation bottleneck. In this work, we propose an accelerator design methodology - Frequency Compensated Memory Packing (FCMP) - which improves the OCM utilization efficiency of dataflow accelerators with minimal reduction in throughput and no modifications to the physical structure of FPGA OCM. To validate our methodology, we apply it to several realizations of medium-sized CIFAR-10 inference accelerators and demonstrate up to 30% reduction in OCM utilization without loss of inference throughput, allowing us to port the accelerators from Xilinx Zynq 7020 to 7012S, reducing application cost. We also implement a custom dataflow FPGA inference accelerator for a quantized ResNet-50 CNN, utilizing on-chip weights, the largest topology ever implemented with this accelerator architecture. We demonstrate that by applying FCMP to the ResNet accelerator, the OCM bottleneck is alleviated which enables the accelerator to be ported from Alveo U250 to the smaller Alveo U280 board with less throughput loss compared to alternative techniques. By providing a finer-grained trade off between throughput and OCM requirements, FCMP increases the flexibility of custom dataflow CNN inference designs on FPGA.

I. INTRODUCTION

The FPGA implementation of Convolutional Neural Network (CNN) inference accelerators has become a hot topic of research as the efficiency of the inference process increasingly drives the cost of ML-based mobile and datacenter workloads [1], [2]. Modern high-accuracy CNNs for vision applications have deep and complex topologies, consisting of a multitude of convolutional layers [3] [4]. While these deep networks have well-established advantages with regard to expressiveness and generalization [5], [6], they create difficulty for FPGA acceleration due to the large number of parameters which need to be stored and operations to be computed for each inference.

Figure 1 illustrates two approaches to inference acceleration on FPGA. Most FPGA inference accelerators are based on overlay architectures (Fig. 1 left) [7], [8], i.e. general-purpose matrix multiplication circuits onto which compute is scheduled to execute the CNN layers in sequence. This approach is flexible, as it enables potentially any CNN topology to be executed by a single accelerator, but is not efficient: it requires frequent transfers of weights and activations (i.e. outputs of hidden layers) between FPGA on-chip scratchpad memories, and external memory (DDR/HBM), arithmetic precision is

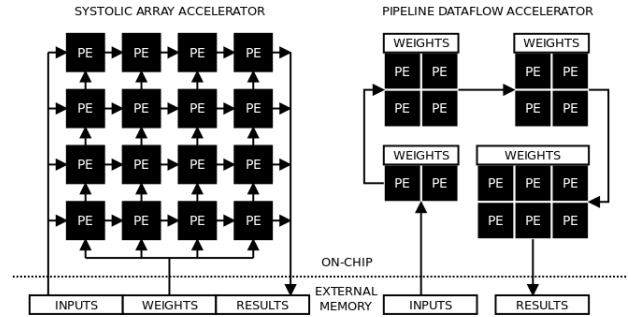


Fig. 1: FPGA Inference Accelerator Architectures

fixed, and the high execution latency of layer-by-layer compute makes real-time inference difficult.

An alternative FPGA accelerator architecture is custom dataflow (Fig. 1 right), whereby the structure of the FPGA-implemented logic mirrors the topological structure of the CNN, creating a pipeline of smaller compute units, each performing the computation corresponding to one specific CNN layer using weights stored in On-Chip Memory (OCM) and the arithmetic of processing elements is tailored to the precision requirements of each layer. As weights and activations never move off-chip, latency and power dissipation are reduced. Custom dataflow for CNN inference has achieved the lowest latency, highest throughput, and lowest power dissipation for image classification and related tasks up to CIFAR-10 complexity [9]. However, this accelerator architecture cannot scale to arbitrarily large CNNs, as it is fundamentally limited by available on-chip resources - LUTs and DSP blocks - required to implement compute units for each CNN layers, and critically, the size of OCM required to store the weights. Furthermore, due to mismatches between the sizes of weight buffers and of FPGA OCM blocks, a significant part of the available OCM cannot be utilized for storing weights, further limiting the size of networks which can be accelerated in custom dataflow.

In this work we approach this problem and propose a method to make more of the FPGA OCM available for weight storage in custom pipeline dataflow (onwards denoted simply as dataflow). We present the largest and deepest CNNs implemented to date in single-chip dataflow on FPGA. Using quantization techniques, we develop high-accuracy CNN image classification models based on the ResNet-50 topology, utilizing binary (1-bit) and ternary (2-bit) weights respectively, thus enabling their implementation in existing FPGAs.